ITA0608 MACHINE LEARNING LAB

PROGRAMS

Name: Balaji Rohith MV

Reg. No: 192521269

Date: 01/08/2026

Experiment 3: Decision

Tree-based ID3 Algorithm

AIM:

To demonstrate the working of the Decision Tree-based ID3 algorithm using an appropriate dataset and classify a new sample.

PROCEDURE :

1. Import the required libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Create a Decision Tree classifier using the Entropy criterion (ID3).
5. Train the classifier using the training data.
6. Test the model using the test data.
7. Predict the class label for a new sample.
8. Display the accuracy and prediction results.

ALGORITHM :

1. Start.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Select the attribute with the highest Information Gain.
5. Create the root node using the selected attribute.
6. Repeat the process recursively for each branch until all samples belong to the same class or no attributes remain.
7. Build the Decision Tree.
8. Use the trained tree to classify a new sample.
9. Display the prediction and accuracy.
10. Stop.

PROGRAM CODE :

```python
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

data = load_iris()

X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = DecisionTreeClassifier(
    criterion="entropy",
    random_state=42
)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

new_sample = [[5.1, 3.5, 1.4, 0.2]]
prediction = model.predict(new_sample)

print("Predicted Class for Test Data:")
print(y_pred)

print("Accuracy:", accuracy_score(y_test, y_pred))

print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("New Sample Prediction:", data.target_names[prediction[0]])
```

OUTPUT :

```
Predicted Class for Test Data:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
Accuracy: 1.0
Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
New Sample Prediction: setosa
```

RESULT :

The Decision Tree-based ID3 algorithm was successfully implemented using the Iris dataset. The decision tree correctly classified the test data and accurately predicted the class of the new sample.